META

Meta Stock Time-Series Analysis

Time-Series Modeling, Market Events, and Forecasting | 2021–2026

AN 306 – Business Intelligence & Data Visualization
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Project roadmap

WORKFLOW

1	**Stationarity & ARIMA**	ADF, differencing, ACF/PACF
2	**Returns, Outliers & Trends**	Returns, volume, moving averages
3	**Forecasting**	SARIMAX vs. AR(30), backtesting
4	**Market Events**	Candlestick, annotations, support/resistance
5	**Financial Story & Conclusions**	Link events to statements and conclusions

Step 1 — Stationarity

META's raw closing-price series was non-stationary; first differencing achieved stationary.

ADF RESULTS

0.8707

Raw closing-price p-value

0.0000

First-differenced p-value

d = 1

ARIMA differencing term

Interpretation

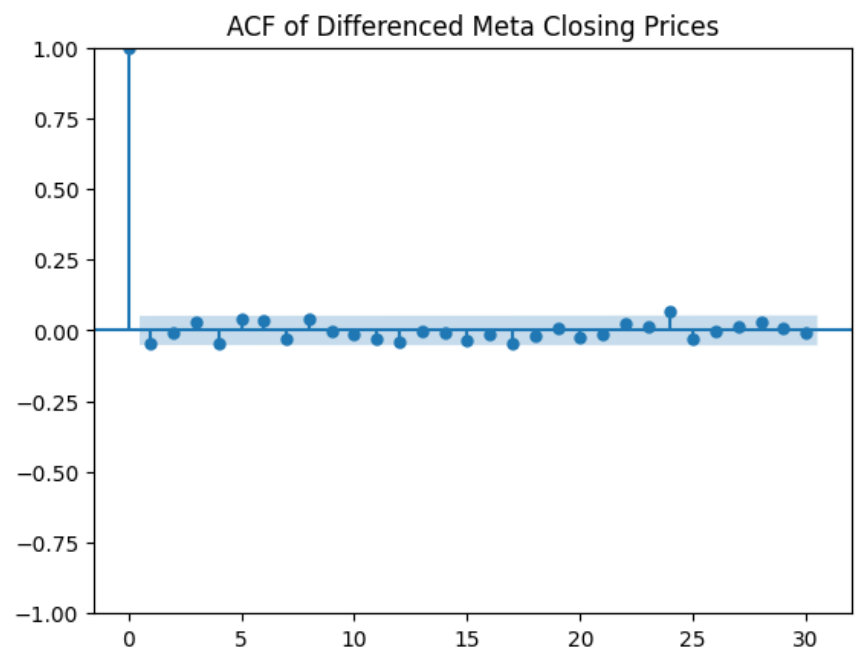
- Raw META prices were non-stationary because the ADF p-value was greater than the 0.05 significance level.
- After first differencing, the p-value fell to 0.0000, indicating stationarity.
- Because one difference was enough, the ARIMA differencing parameter is d = 1.

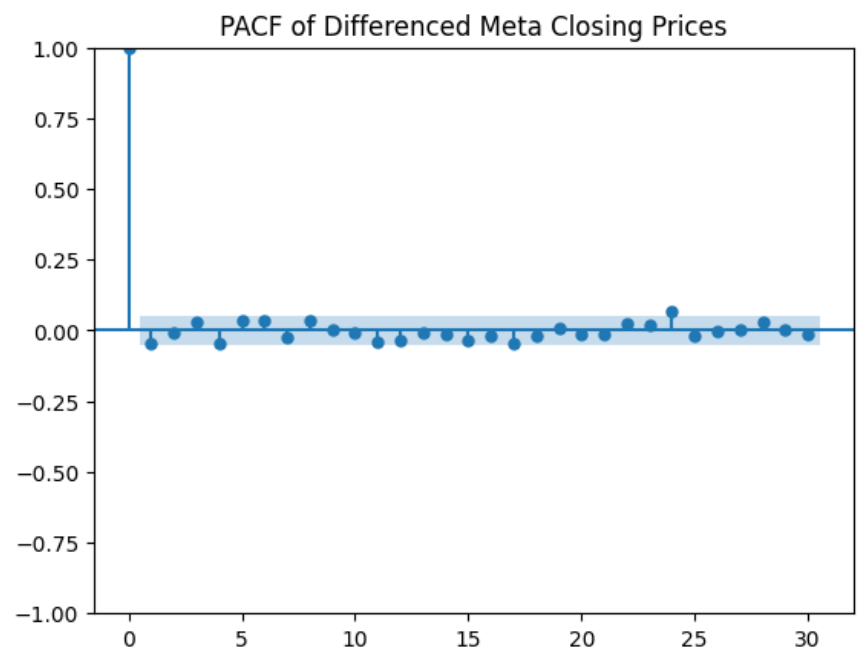
ADF statistic: −0.6014 → −39.8351 after first differencing

Step 1 — ACF/PACF and ARIMA setup

Both plots are mostly flat after differencing, suggesting little meaningful lag dependence.

MODEL FOUNDATION





Starting specification: ARIMA(0,1,0)

ACF → q ≈ 0 • PACF → p ≈ 0 • Compare nearby specifications during forecasting

Step 2 — Volatility and outliers

The 3-standard-deviation rule reduced 72 ±5% trading days to 17 statistically extreme outliers.

DAILY RETURN PROFILE

+0.106%	2.738%	+8.32%	−8.11%
Mean daily change	Standard deviation	Upper 3-SD boundary	Lower 3-SD boundary

17

3-SD outlier days

Extreme daily moves beyond the thresholds above.

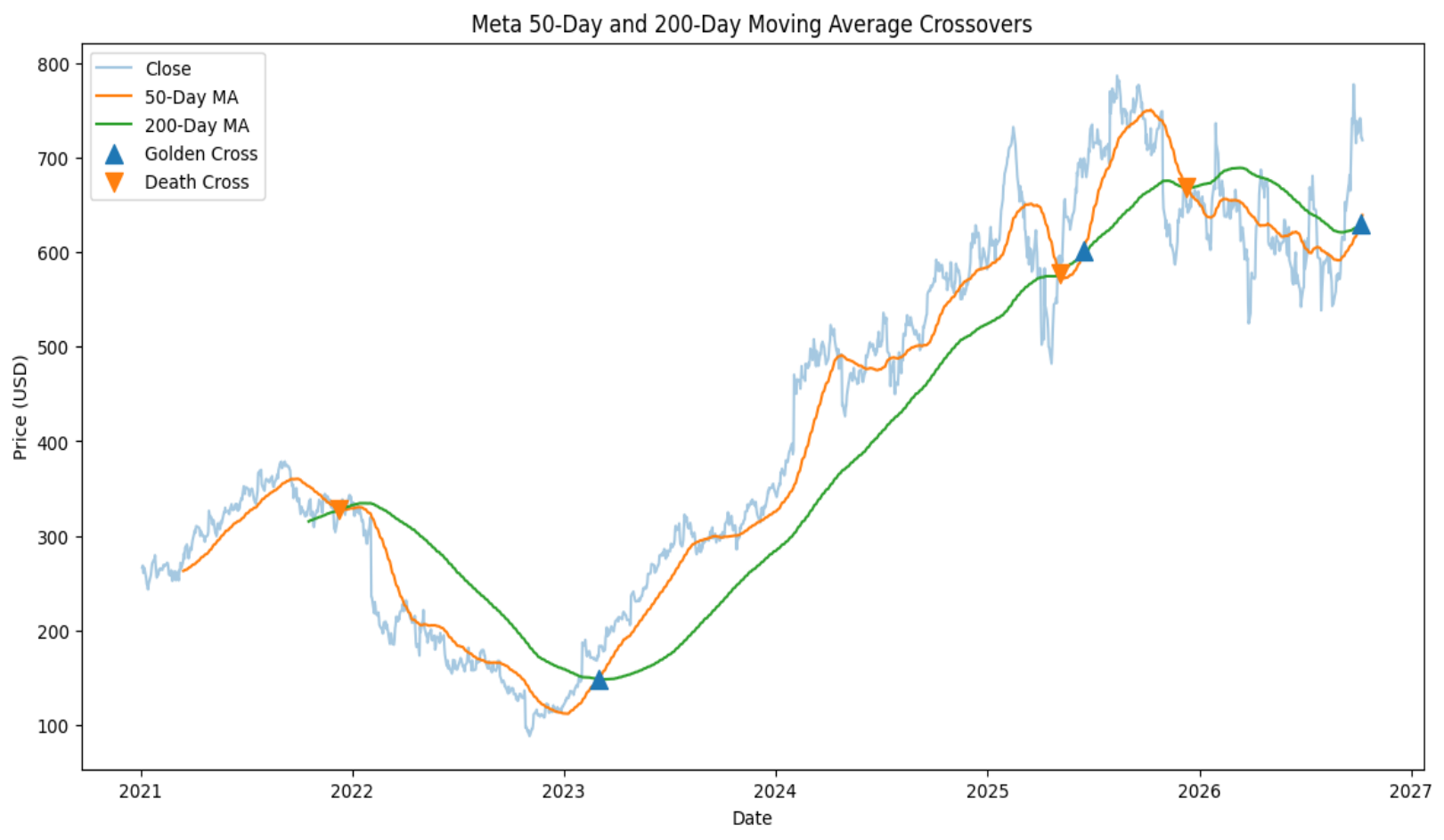
Key interpretation

- The 3-SD rule isolates the most statistically unusual daily price movements.
- These 17 outliers identify the strongest candidates for explaining major market reactions.
- Extreme price moves are especially meaningful when paired with unusually high trading volume.

Step 2 — Moving averages and trend signals

META experienced several long-term momentum reversals, reflected by three Golden Crosses and three Death Crosses.

TREND ANALYSIS



Golden Crosses

Mar 2, 2023
Jun 16, 2025
Oct 6, 2026

Death Crosses

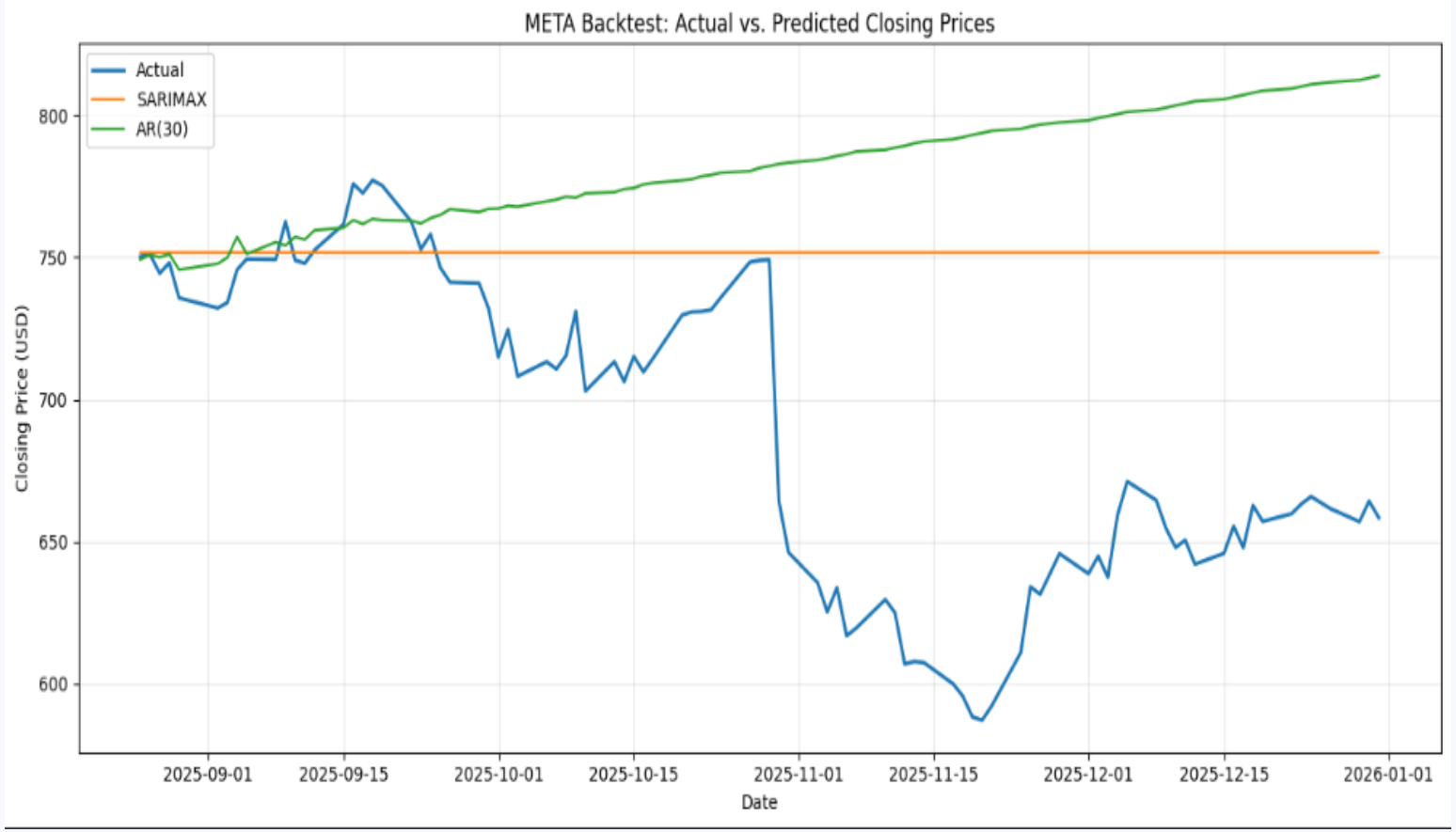
Dec 10, 2021
May 6, 2025
Dec 10, 2025

These crossovers indicate changes in momentum, not guaranteed future price direction.

Step 3 — SARIMAX outperformed AR(30)

Backtesting showed that SARIMAX produced lower prediction error for META.

MODEL COMPARISON



MODEL RESULTS

SARIMAX

MAE	RMSE
64.18	82.37

AR(30)

MAE	RMSE
91.80	114.94

Lower error = better fit

Preferred model: SARIMAX

Key insight: SARIMAX was closer to the actual price path overall, even though both models missed the sharp late-2025 decline.

Step 3 — META 90-trading-day forecast

SARIMAX projects a relatively flat path while uncertainty widens quickly over the forecast horizon.

FINAL FORECAST



Takeaway: the point forecast is stable, but the widening interval shows that confidence falls quickly as the horizon extends.

KEY FINDINGS

Flat forecast

Predicted prices stay near META's recent level instead of showing a strong up or down trend.

Widening uncertainty

The 95% confidence interval expands substantially as the forecast moves farther into the future.

Model limitation

Historical prices cannot anticipate earnings surprises, regulation, company news, or macroeconomic shocks.

No strong directional signal

Step 4 — Historical Stock Performance

META's 2022 decline was followed by a sustained recovery and renewed volatility.

TREND ANALYSIS



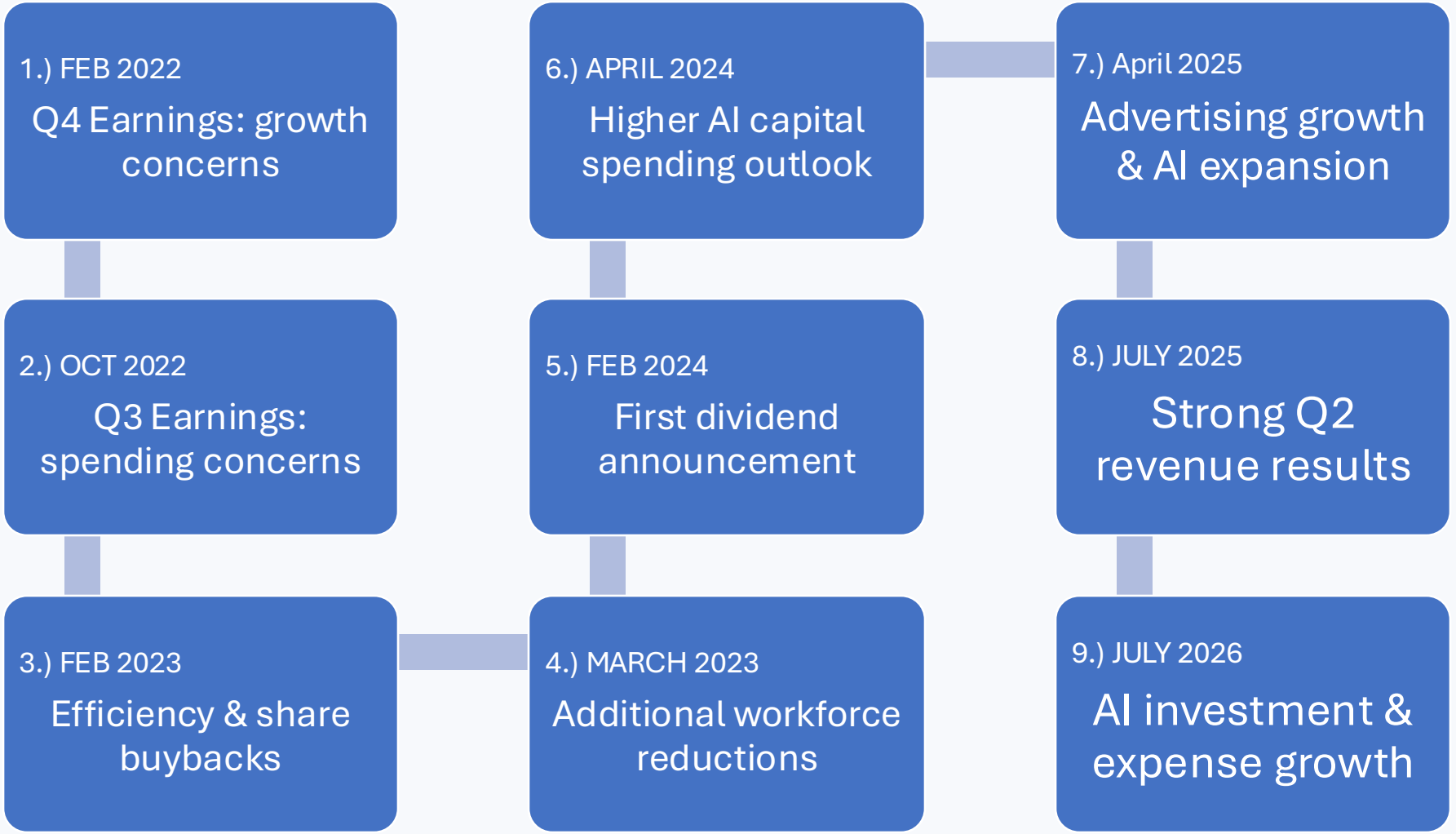
Click here to explore the interactive chart

Key Insight ~ Major turning points reflect shifts in growth, spending, and profitability expectations.

Step 4 — Events behind the price movement

Nine company announcements connect historical volatility to business decisions.

MARKET EVENT TIMELINE



Step 4 — Support and resistance

Recent price reversals suggest potential trading zones, not guaranteed price barriers.

TECHNICAL ANALYSIS



KEY PRICE ZONES

SUPPORT

578.79 USD

3 historical turning points

RESISTANCE

742.73 USD

4 historical turning points

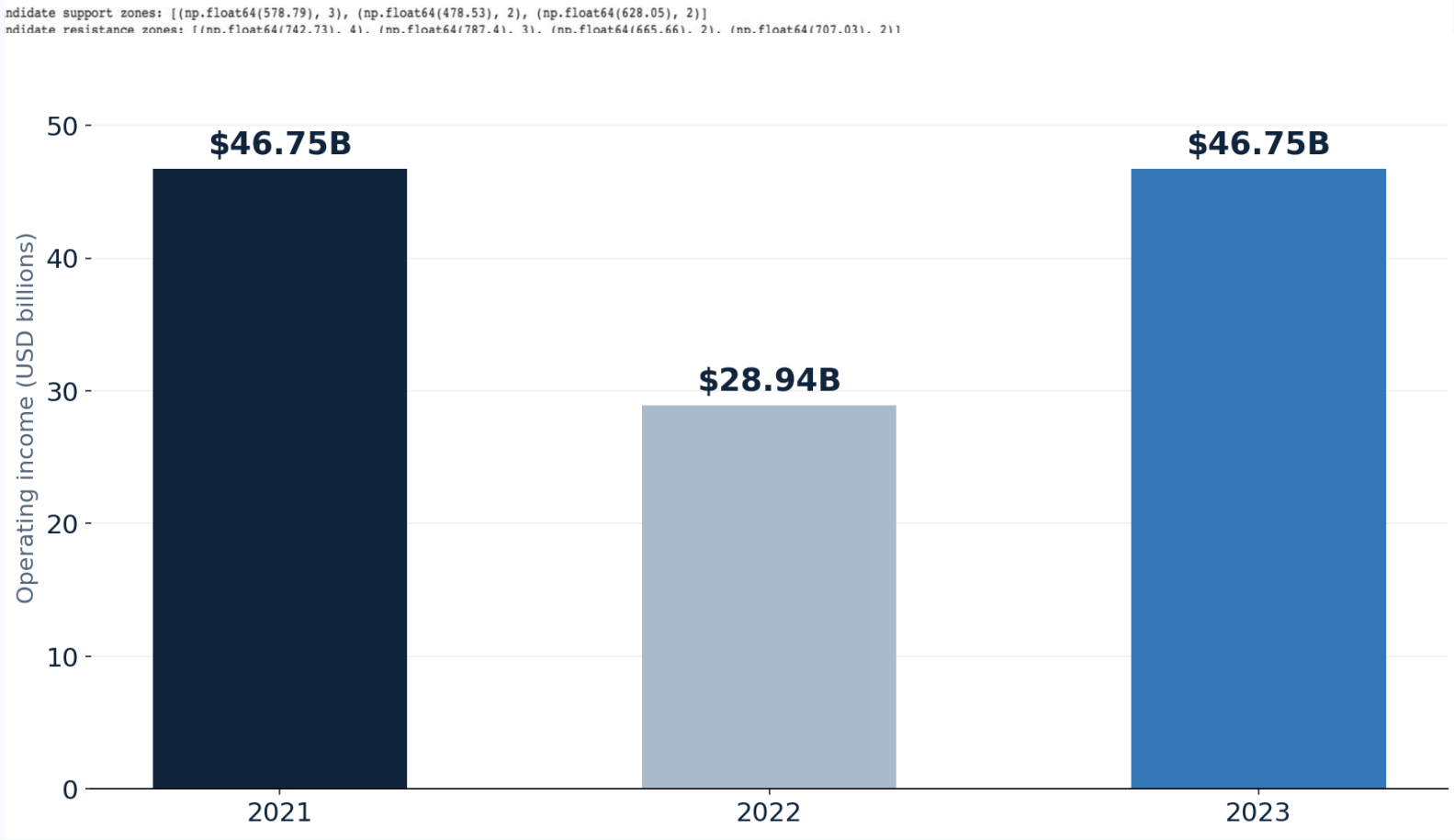
LATEST CLOSE

658.40 USD

Step 5 — Meta’s financial turnaround

Operating income fell in 2022, then recovered as Meta prioritized efficiency.

FINANCIAL STATEMENT ANALYSIS



OPERATING INCOME

2021 Baseline

$46.75B

-strong pre-decline profitability

2022 Decline

$28.94B

~38% below 2021

2023 Rebound

$46.75 B

Step 5 — AI investment and Accounting

AI Infrastructure spending affects cash flow immediately, but earnings overall over time.

CAPITAL SPENDING | STATEMENT LINK

1

CASH FLOW STATEMENT

Cash spent on servers, data centers and infrastructure

Investing cash outflow

2

BALANCE SHEET

Qualifying infrastructure is capitalized as assets

Property & equipment rises

3

INCOME STATEMENT

Capitalized assets are depreciated over useful lives

Future operating expenses rise

INVESTMENT TRADE-OFF

CASH TODAY

Outflow
Infrastructure Spending

ASSETS CREATED

CapEx
Recorded on Balance Sheet

EARNINGS LATER

D&A

Step 5 — Outlook and Conclusion

Statistical forecasts are a baseline; business developments can shift the path.

MODEL LIMITATIONS | BUSINESS OUTLOOK

01

ADVERTISING

Does revenue growth continue?

Core earnings engine

02

AI INVESTMENT

Do investments produce returns?

Spending versus payoff

03

PROFITABILITY

Can operating margins hold?

Costs and depreciation

FORECAST CONTEXT

BEST TESTED MODEL

SARIMAX
Lower error than AR(30)

90-DAY OUTLOOK

Flat
Uncertainty widens over time

KEY LIMITATION

Events